Timing, Not Transform: Shortcut-Controlled and Externally Validated Murmur Detection on CirCor

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***Abstract*—We asked which part of a classical phonocardiogram (PCG) pipeline carries murmur information on the CirCor DigiScope data, using pre-registered contrasts, subject-grouped cross-validation in WEKA and a fixed decision rule per hypothesis. *(i) Cardiac-cycle timing matters:* keeping the same band features but destroying their alignment to the expert S1/systole/S2/diastole labels costs +0.131 to +0.174 patient AUROC (H5a–c; all 25 folds). *(ii) The time–frequency transform does not:* wavelet packets (WPD db4) are never better than matched STFT bands (differences −0.042 to +0.003), and wavelet scattering does not beat them either (H11a/b not supported). *(iii) External validation:* a whole-signal STFT model trained once on all CirCor patients reaches AUROC 0.709 [0.675, 0.741] for congenital heart disease (CHD) on the independent paediatric ZCHSound cohort, with the physiological subtype order PDA > VSD > ASD > PFO (H10a/b supported). An automatic segmentation pipeline with a learned span detector matches expert-label performance (H9a), but its advantage over a whole-signal STFT baseline that is given the same span information is not established (H12a: +0.046 [−0.001, +0.093]). A test-time oracle changes what a number measures: in our data the expert segmentation is worth up to 0.106 AUROC over a naive automatic pipeline.**

I. Introduction and related work

Automated murmur detection from PCG has been driven by the George B. Moody PhysioNet Challenge 2022 and the CirCor DigiScope dataset [1], [2]. Many pipelines segment each recording into cardiac cycles and compute time–frequency features per phase; the CirCor release includes expert S1/systole/S2/diastole annotations (`.tsv`) that make this easy. Two questions are rarely separated: is it the *timing* (alignment to the cardiac cycle) or the *transform* (e.g. wavelets) that helps, and how much of a reported number depends on expert information that a deployed system would not have? Following Cao et al. [3], we call a test-time use of expert annotations *oracle conditioning* and report the gap between oracle and estimated conditioning.

Related work. Vimalajeewa et al. [4] (B1) proposed multi-scale wavelet features for murmur detection on CirCor; we re-implement their features under our folds. The winning Challenge entry used an RNN with parallel hidden semi-Markov models [5]; HSMM segmentation follows Springer et al. [6]. Kutlu and Çolak [7] already test CirCor-trained murmur classifiers zero-shot on PhysioNet/CinC 2016 and Yaseen 2018, so external validation of CirCor-trained models is not new; ZCHSound [11] was not among their targets. Ladislaus et al. [8] show on CirCor that aggregation rules and metric conventions change measured noise robustness, a point related to ours about test-time conditioning. Nogueira et al. [9] and Nie et al. [10] segment test data from the expert annotations (Section IV-G). [Conditional on the pending G3b full-text checks finding no other CirCor→ZCHSound evaluation:] to our knowledge this is the first evaluation of CirCor-trained murmur models on ZCHSound, an independent paediatric CHD cohort, with a pre-registered physiology (subtype-ordering) check.

Literature audit (G3b).

> G3b result to be inserted after coding (Part 1d): proportion of included papers with test-time expert segmentation, test–retest Cohen's κ per field, and the headline values by group. Random sample of 30 records (amendment A1), coded by the single author, 10 re-coded blind (amendment A2).

II. Data

CirCor DigiScope v1.0.3 training set (PhysioNet): 942 patients, 3,163 recordings at 4 kHz from up to four auscultation sites, with murmur labels Present/Absent/Unknown and expert segmentation. We use the binary task Present vs Absent (Unknown excluded): 874 patients (179 Present, 695 Absent) from 816 subjects; linked visits share a subject ID. The CirCor hidden test set is not public and was not used. **ZCHSound** [11] (Zhejiang University Children's Hospital; GitHub `WeiJieOvO/ZCHSound-Dataset`, CC BY 4.0): 1,259 participants, one recording each, labelled by echocardiography-confirmed diagnosis (CHD subtypes or NORMAL), split by the authors into a clean set (941 children, 408 with CHD) and a noisy neonatal set (318 children aged five days or less). Its labels are CHD diagnoses, *not* murmur labels. ZCHSound was used for no training, tuning or threshold choice.

III. Methods

Preprocessing (fixed, training-independent): resampling to 2 kHz (polyphase FIR), zero-phase Butterworth band-pass 25–900 Hz (order 4), per-recording z-score. **Representations.** WPD: depth 4, db4 (fixed a priori; db6, sym8 and coif3 as sensitivity only), periodisation, frequency-ordered nodes, giving 16 bands of 62.5 Hz on an 8 ms grid. STFT bands are matched to the same 16 band edges, time grid and dimensionality, so wavelet vs Fourier is a like-for-like contrast. *wpds* is WPD with the STFT's Hann window applied as temporal smoothing. Wavelet scattering (G4): kymatio [13] `Scattering1D(J=6, Q=(8,1))` at 2 kHz, first order (scat1) and first+second order (scat12) [12], aggregated into the same bands. **Features.** Whole-signal (W): log relative band energies, burstiness $\log(q_{90}/q_{50})$ per band, two entropies.

Cycle-aligned (A, E2, E4): per band log(systolic/S1 energy) and log(systolic/diastolic energy), two entropies; a recording contributes aligned features only with enough annotated systole, diastole and S1. Patient vector: median and maximum over the patient's recordings. Baselines: B0 (MFCC and spectral descriptors) and B1 [4]. **Segmentation.** *Expert* labels (oracle). Controls with timing destroyed: *perm* (labels permuted within a recording) and *shift* (circular shift). Automatic: E1 envelope-peak segmenter; E1b LR-HSMM [6] on the whole recording; E4 = HSMM plus a learned span detector (logistic regression on six window features) that keeps 1-s windows likely to be annotatable, trained on training folds only. Diagnostics D1–D3 use the expert spans at test time and are labelled as such. **Shortcut control.** E4's selected-span fraction and missingness predict the class on their own (span fraction alone: AUROC 0.646). S1 adds these shortcut variables (SC) to whole-signal STFT. G2 defines E4q, which keeps in every recording the same top-q fraction of windows (q from the training fold), removing the shortcut by construction, and compares it with W_stft + SC_q (H12a), with random windows of the same fraction (rand-q, H12b) and an SNR gate (snr-q, H12c). **Classifiers and evaluation.** WEKA 3.9.7 [14]: RandomForest [15] (`-I 300 -K 0 -S 1 -num-slots 1`, primary) and SMO [16] (linear PolyKernel, logistic calibration, secondary), called in batch from Python; WEKA's own row-wise cross-validation is never used because it would split a patient's recordings across folds. Folds: 5 repeats $\times$ 5 subject-grouped outer folds, fixed before any result (sha256 `c4aa74c7...2339`). AUROC is computed on repeat-averaged patient scores, with a subject bootstrap (2000 resamples) for 95 % CIs. Thresholds come from inner training folds only (specificity ≥ 0.80). **Decision rules** (pre-registered per hypothesis): typically $\Delta \geq 0.03$, CI lower bound > 0 and $\geq 18/25$ folds with $\Delta > 0$; non-inferiority $\Delta \geq -0.03$ for H6a/H8a/H9a. **Timeline.** Phase 1 (v1, Linux) was locked before any result; Phase 2 experiments E1b, D1–D3, E4 and S1 were each pre-registered after the preceding results; Phase 3 (G1a, G2, G3, G4) was pre-registered after all Phase 1–2 results (Appendix B).

IV. Results

A. Baselines and the main models

Table I lists the RF AUROCs and operating points of the reference models. Cycle-aligned WPD beats whole-signal WPD (H3: +0.113 [+0.065, +0.165]) and B1 (H4b: +0.072 [+0.027, +0.122]), but not B0 (H4a: +0.051 [−0.002, +0.102]).

B. Finding 1: timing, not span choice

With expert labels, destroying timing costs +0.131 to +0.174 AUROC (perm, H5a–c) and +0.192 to +0.206 (shift, H5d); all are supported in 25/25 folds. The diagnostics split this exactly (Fig. 1): within-span timing is worth +0.169 to +0.193, while span selection without timing contributes −0.062 to −0.020. Automatic segmentation alone fails: the LR-HSMM is near expert timing inside expert spans (expert − D1: +0.010 to +0.015), but on whole recordings it loses +0.077 to +0.091 through span selection (H8a: −0.091 [−0.135, −0.045]). E4's learned span detector closes this gap (H9a, non-inferior to expert labels: −0.007 [−0.044, +0.031] for wpd) and beats B1 (H9b: +0.065 to +0.083).

TABLE I
Main models (RF, patient level; operating point: inner-fold threshold for specificity ≥ 0.80; W.acc = Challenge weighted accuracy at that threshold).

Model (RF)	AUROC [95 % CI]	Se	Sp	W.acc
B0 acoustic baseline	0.747 [0.699, 0.793]	0.573	0.815	0.679
B1 Vimalajeewa et al.	0.726 [0.677, 0.773]	0.551	0.813	0.666
whole-signal STFT (W_stft)	0.751 [0.701, 0.794]	0.577	0.808	0.678
expert-aligned STFT (oracle)	0.815 [0.772, 0.853]	0.663	0.810	0.727
E4 automatic STFT	0.809 [0.766, 0.849]	0.679	0.798	0.731
E4q shortcut-free STFT	0.799 [0.756, 0.842]	0.637	0.810	0.713

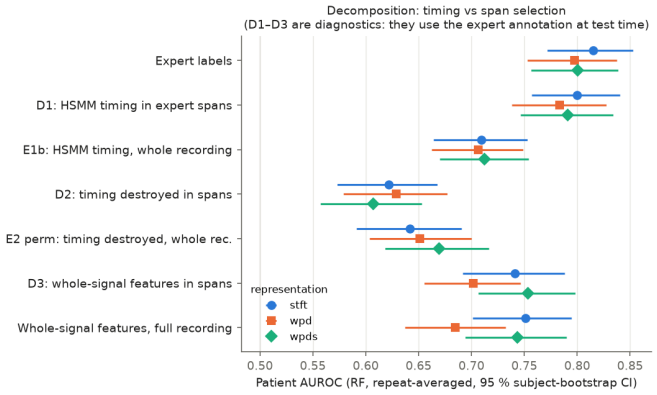


Fig. 1. Decomposition (RF AUROC, 95 % CI). D1–D3 use the expert annotation at test time (diagnostics).

C. Finding 2: the transform does not matter

Under every labelling, WPD and smoothed WPD are at most equal to matched STFT bands (Fig. 2; H2a: −0.017 [−0.034, 0.000], not supported), and equivalence within ± 0.03 is not shown either (−0.015 [−0.032, +0.003]). Wavelet scattering (G4, Fig. 3) does not beat matched STFT bands: H11a −0.026 [−0.058, +0.005], H11b −0.013 [−0.044, +0.017]. Second-order paths add +0.013 [+0.001, +0.025], below the pre-registered 0.03 (H11c not supported).

D. Automatic pipeline and the shortcut check

Against plain whole-signal STFT, the H9c rule is met for stft (+0.058 [+0.011, +0.108]) but not for the pre-registered primary wpd (+0.040 [−0.003, +0.089]); this comparison does not yet account for the span shortcut. Once W_stft is given the span/missingness shortcut variables (S1), E4's margin is +0.026 to +0.043 with CIs including zero ("possibly shortcut-driven"). E4q removes the shortcut: the shortcut variables alone drop to AUROC 0.482 [0.434, 0.529], the share of test recordings without aligned features falls from 15.8 % (Absent) and 9.9 % (Present) under E4 to 0.6 % and 0.8 %, and AUROC is essentially unchanged (E4q − E4: −0.009

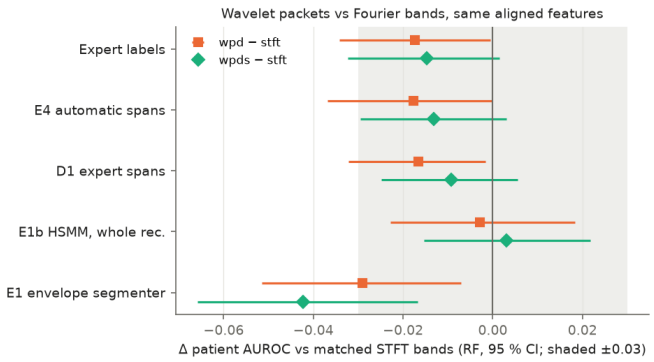


Fig. 2. Wavelet packets minus matched STFT bands, same aligned features (RF, 95 % CI; shaded band ±0.03).

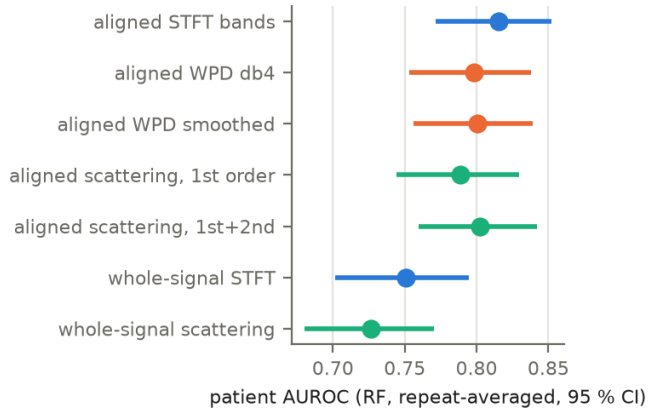


Fig. 3. G4: matched STFT, WPD and scattering features (RF AUROC, 95 % CI).

[−0.041, +0.019]). Learned selection beats random windows (H12b: +0.066 [+0.030, +0.100]). The primary contrast H12a, E4q − (W_stft + SC_q), is +0.046 [−0.001, +0.093] with 23/25 folds positive: the CI lower bound is below zero, so H12a is *not established* ("possibly shortcut-driven").

E. Finding 3: external validation on ZCHSound

Models were trained once on all 874 CirCor patients (whole-signal features, since ZCHSound has one recording and no annotation) and applied unchanged (Fig. 4, Table II). On the clean set, W_stft reaches AUROC 0.709 [0.675, 0.741] (H10a supported; internal CirCor 0.751); at the CirCor threshold, sensitivity 0.520 and specificity 0.786. By subtype vs NORMAL: PDA 0.854 [0.796, 0.905], VSD 0.785 [0.744, 0.823], ASD 0.659 [0.605, 0.716], PFO 0.522 [0.447, 0.596]; the pre-registered order holds (H10b: VSD − PFO +0.263 [+0.180, +0.343], VSD − ASD +0.126 [+0.064, +0.184]). On the noisy neonatal set every model is at chance (W_stft 0.535 [0.471, 0.597]). A post-hoc age check (not pre-registered) shows that age alone separates CHD from NORMAL (AUROC 0.231), but the W_stft score is unrelated to age (Spearman −0.028) and its AUROC within age tertiles is 0.694, 0.774 and 0.745.

TABLE II
EXTERNAL AUROC [95 % CI] FOR ANY CHD (RF). B0 ENTERS NO VERDICT (ARTEFACT, SEE TEXT).

Model (RF, trained on CirCor)	clean	noisy	CirCor internal
W_stft (pre-registered, H10a)	0.709 [0.675, 0.741]	0.535 [0.471, 0.597]	0.751
W_wpd_db4_smooth	0.784 [0.751, 0.813]	0.497 [0.436, 0.562]	0.743
W_scat1	0.559 [0.521, 0.597]	0.411 [0.347, 0.471]	0.727
B0 (artefact, no verdict)	0.699 [0.662, 0.732]	0.503 [0.438, 0.570]	0.747

TABLE III
CONDITIONING GAPS (RF AUROC DIFFERENCES, 95 % CI).

Gap (RF AUROC)	stft	wpd	wpds
Δ_{or} expert − perm	0.174 [0.115, 0.229]	0.147 [0.094, 0.199]	0.131 [0.079, 0.185]
Δ_{or} expert − shift	0.202 [0.152, 0.254]	0.192 [0.141, 0.248]	0.206 [0.153, 0.260]
Δ_{oe} expert − E1	0.050 [0.011, 0.090]	0.062 [0.021, 0.102]	0.077 [0.037, 0.120]
Δ_{oe} expert − E1b	0.106 [0.063, 0.148]	0.091 [0.045, 0.135]	0.088 [0.045, 0.130]
Δ_{oe} expert − E4	0.006 [−0.031, 0.045]	0.007 [−0.032, 0.044]	0.005 [−0.035, 0.044]
Δ_{oe} expert − E4q	0.016 [−0.023, 0.055]	0.006 [−0.034, 0.047]	0.012 [−0.029, 0.053]

F. Conditioning gaps (G3a)

Table III gives the oracle–estimated gaps Δ_{oe} and the oracle–random gaps Δ_{or} [3]. Replacing the expert segmentation by a naive automatic one costs 0.050 [0.011, 0.090] (E1) to 0.106 [0.063, 0.148] (E1b) AUROC for stft; with E4 and E4q the gap is 0.006 [−0.031, 0.045] and 0.016 [−0.023, 0.055].

G. Comparison with recent work

B1. Against the closest published classical method [4], re-implemented under our folds, E4 is better by +0.083 [+0.036, +0.135] (stft, 24/25 folds) and cycle-aligned WPD by +0.072 [+0.027, +0.122] (H4b). **Case study.** Nogueira et al. [9] report an "overall accuracy" of 0.965, defined as (Se+Sp)/2 (their Eq. 3, p. 9), for patients whose test segments start at expert-annotated S1 onsets (their §3.2, p. 4), in a ten-fold patient-level CV of the 735 patients with all four sites (§2, §4). The metric is a balanced accuracy at one threshold, not an AUROC, and the label is described as "normal vs abnormal". On the same balanced-accuracy scale, our expert-label model reaches 0.736 at the inner-fold specificity-0.80 threshold, E4 0.739 and E4q 0.724. Nie et al. [10] report 97.49 % snippet-level accuracy on 5-s snippets taken only from annotated periods (their §2, §4), and note themselves that this differs from recording-level Challenge evaluation (§5). These numbers are not on the same scale as ours: different metric, unit and cohort, and test-time conditioning on the expert annotation, which in our data is worth up to 0.106 AUROC over a naive automatic pipeline.

V. DISCUSSION AND LIMITATIONS

- **H12a is not established** (+0.046 [−0.001, +0.093]); a deployable gain of the timing pipeline over a shortcut-aware whole-signal STFT baseline is not shown.
- **Chance on noisy recordings:** all models fail on the noisy neonatal ZCHSound set.
- **G4 is negative:** scattering adds nothing over matched STFT bands.
- **ZCHSound labels are CHD diagnoses, not murmurs;** silent defects (PFO) are at chance, as expected.

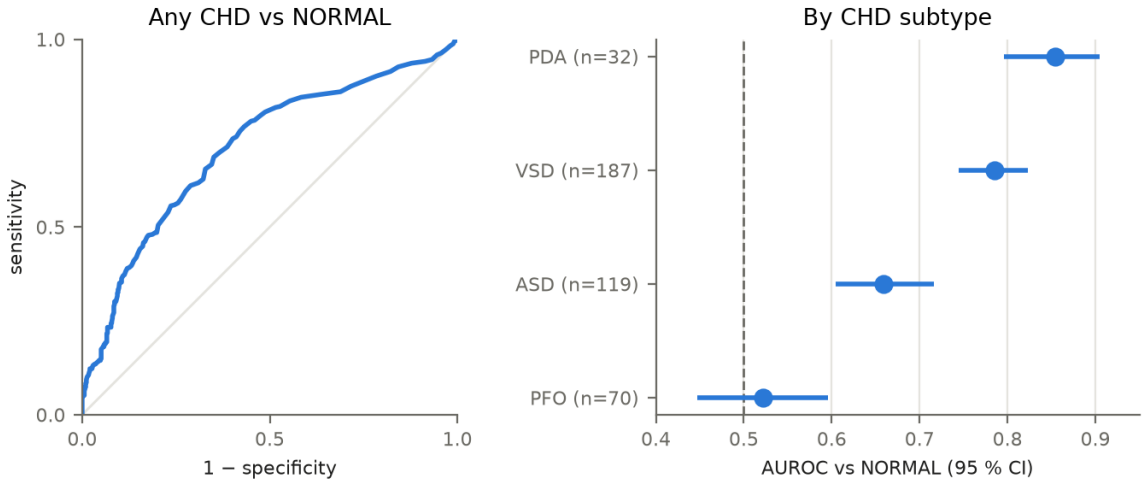


Fig. 4. ZCHSound clean set, RF W_stft trained on CirCor: any-CHD ROC and AUROC by subtype vs NORMAL (95 % CI).

- **No CirCor hidden test set;** all CirCor results are cross-validated on the public training set.
- **The Unknown class is excluded,** so the task is easier than the three-class Challenge task.
- **Sequential design:** E1b, D1–D3, E4, S1 and Phase 3 were designed after earlier results on the same folds; ZCHSound is the only fully independent evaluation.
- **SMO is unstable on scattering features:** with features identical to 10^{-9} across platforms, SMO scores moved by up to 0.097 (CV) and 0.158 (ZCHSound), against 0.017 for RF; no SMO scattering AUROC is used as evidence.
- **Single coder** for the literature audit: reliability is intra-rater test–retest only (amendment A2).
- **B0 artefact:** B0 SMO reaches 0.834 on ZCHSound, but B0 RF trained on pulmonary-site recordings is far below chance (0.260); the B0 external signal reflects recording characteristics, not murmur.

VI. Conclusion and Future Work

On CirCor, cardiac-cycle timing carries the murmur information and the choice of time–frequency transform does not; a whole-signal model transfers to an independent paediatric CHD cohort with the expected physiological ordering. Next steps: signal-quality gating (the noisy set), learned models evaluated under the same subject-grouped and shortcut-controlled protocol, and a murmur-labelled external set.

Reproducibility

Code, protocols and saved outputs: private repository `pm2m1/ec418-circor-research`. Tags `v1.0-pilot` (Phase 1), `protocol-v2-lock`, `results-final` (Phase 2) and `phase3-results` (Phase 3); the Phase 3 pre-registration tag `protocol-v3-prereg` lives in the separate cloud-sandbox history, verified in `results/logs/p3_mac_provenance.txt`; manifests `results/v1_freeze/MANIFEST_v1.sha256` and `results/final_freeze/MANIFEST_final.sha256`. Phase 1 ran on Linux and was reproduced on macOS (`docs/REPRO_MAC.md`); every number in this report is generated from saved tables by `scripts/p4_numbers.py` and checked by `scripts/p4_check_numbers.py`.

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APPENDIX A
EVERY PRE-REGISTERED HYPOTHESIS

Table IV lists every pre-registered hypothesis with its result (RF), including those not supported.

APPENDIX B
PRE-REGISTRATION TIMELINE

v1 protocol and folds locked before any result (tag `v1.0-pilot` after the pilot); Phase 2 locked after v1 (tag `protocol-v2-lock`); E1b, D1–D3, E4 and S1 each pre-registered (tags `e1b-prereg`, `diag-prereg`, `e4-prereg`, `s1-prereg`) after the preceding results; Phase 3 pre-registered after all Phase 1–2 results (tag `protocol-v3-prereg` in the cloud-sandbox history, amendments A1–A3). Full commit-level timeline: `docs/RESULTS_SUMMARY.md`.

APPENDIX C
PIPELINE

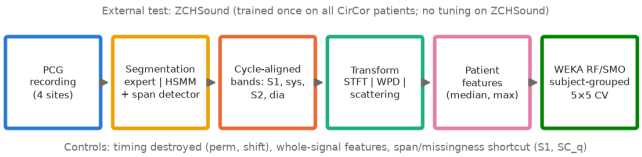


Fig. 5. Pipeline and controls.

TABLE IV
ALL PRE-REGISTERED HYPOTHESES (RF, REPEAT-AVERAGED PATIENT AUROC, SUBJECT-BOOTSTRAP 95 % CI).

Hyp.	Contrast (RF)	Δ AUROC [95 % CI]	folds > 0	verdict
H1	B1+SVM recording AUC: random split - subject-grouped	+0.015 [−0.004, +0.035]	n/a	NOT SUPPORTED
H2a	A_wpd_db4 - A_stft	−0.017 [−0.034, 0.000]	3/25	NOT SUPPORTED
H2b	W_wpd_db4 - W_stft	−0.066 [−0.095, −0.036]	1/25	NOT SUPPORTED
H3	A_wpd_db4 - W_wpd_db4	+0.113 [+0.065, +0.165]	25/25	SUPPORTED
H4a	A_wpd_db4 - B0	+0.051 [−0.002, +0.102]	21/25	NOT SUPPORTED
H4b	A_wpd_db4 - B1	+0.072 [+0.027, +0.122]	24/25	SUPPORTED
H5a	E2_wpd_true - E2_wpd_perm	+0.147 [+0.094, +0.199]	25/25	SUPPORTED
H5b	E2_stft_true - E2_stft_perm	+0.174 [+0.115, +0.229]	25/25	SUPPORTED
H5c	E2_wpds_true - E2_wpds_perm	+0.131 [+0.079, +0.185]	25/25	SUPPORTED
H5d	E2_wpd_true - E2_wpd_shift	+0.192 [+0.141, +0.248]	25/25	SUPPORTED
H5d	E2_stft_true - E2_stft_shift	+0.202 [+0.152, +0.254]	25/25	SUPPORTED
H5d	E2_wpds_true - E2_wpds_shift	+0.206 [+0.153, +0.260]	25/25	SUPPORTED
H6a	E1_wpd_auto - E2_wpd_true	−0.062 [−0.102, −0.021]	3/25	NOT SUPPORTED
H6b	E1_wpd_auto - B1	+0.010 [−0.038, +0.060]	15/25	NOT SUPPORTED
H6c(a)	E1_stft_auto - E2_stft_true	−0.050 [−0.090, −0.011]	1/25	NOT SUPPORTED
H6c(b)	E1_stft_auto - B1	+0.039 [−0.009, +0.088]	20/25	NOT SUPPORTED
H6c(a)	E1_wpds_auto - E2_wpds_true	−0.077 [−0.120, −0.037]	1/25	NOT SUPPORTED
H6c(b)	E1_wpds_auto - B1	−0.003 [−0.051, +0.047]	12/25	NOT SUPPORTED
H8a	E1b_wpd_auto - E2_wpd_true	−0.091 [−0.135, −0.045]	1/25	NOT SUPPORTED
H8b	E1b_wpd_auto - B1	−0.019 [−0.064, +0.029]	6/25	NOT SUPPORTED
H8c(a)	E1b_stft_auto - E2_stft_true	−0.106 [−0.148, −0.063]	0/25	NOT SUPPORTED
H8c(b)	E1b_stft_auto - B1	−0.017 [−0.061, +0.031]	11/25	NOT SUPPORTED
H8c(a)	E1b_wpds_auto - E2_wpds_true	−0.088 [−0.130, −0.045]	0/25	NOT SUPPORTED
H8c(b)	E1b_wpds_auto - B1	−0.013 [−0.057, +0.034]	10/25	NOT SUPPORTED
H9a	E4_wpd - E2_wpd_true	−0.007 [−0.044, +0.031]	10/25	SUPPORTED
H9b	E4_wpd - B1	+0.065 [+0.022, +0.115]	25/25	SUPPORTED
H9c	E4_wpd - W_stft	+0.040 [−0.003, +0.089]	21/25	NOT SUPPORTED
H9a	E4_stft - E2_stft_true	−0.006 [−0.044, +0.031]	8/25	SUPPORTED
H9b	E4_stft - B1	+0.083 [+0.036, +0.135]	24/25	SUPPORTED
H9c	E4_stft - W_stft	+0.058 [+0.011, +0.108]	22/25	SUPPORTED
H9a	E4_wpds - E2_wpds_true	−0.005 [−0.044, +0.035]	9/25	SUPPORTED
H9b	E4_wpds - B1	+0.070 [+0.027, +0.120]	25/25	SUPPORTED
H9c	E4_wpds - W_stft	+0.045 [+0.001, +0.094]	21/25	SUPPORTED
equivalence	E2_wpds_true - E2_stft_true	−0.015 [−0.032, +0.003]	5/25	equivalence NOT shown
H7	soft murmurs (grade I/VI sensitivity)	–	-	NOT RUN (dropped, amendment E3-drop)
S1(a)	E4_stft - S1_Wstft_SC	+0.043 [−0.005, +0.096]	21/25	possibly shortcut-driven
S1(a)	E4_stft - S1_Wwpd_SC	+0.095 [+0.043, +0.149]	23/25	not explained by the span/missingness shortcut
S1(a)	E4_wpd - S1_Wstft_SC	+0.026 [−0.020, +0.075]	21/25	possibly shortcut-driven
S1(a)	E4_wpd - S1_Wwpd_SC	+0.077 [+0.027, +0.131]	24/25	not explained by the span/missingness shortcut
S1(a)	E4_wpds - S1_Wstft_SC	+0.030 [−0.015, +0.081]	18/25	possibly shortcut-driven
S1(a)	E4_wpds - S1_Wwpd_SC	+0.082 [+0.031, +0.137]	23/25	not explained by the span/missingness shortcut
H12a	E4q − (W_stft + SC_q), stft	+0.046 [−0.001, +0.093]	23/25	possibly shortcut-driven
H12a	E4q − (W_stft + SC_q), wpd	+0.038 [−0.010, +0.086]	21/25	possibly shortcut-driven
H12a	E4q − (W_stft + SC_q), wpds	+0.035 [−0.013, +0.083]	20/25	possibly shortcut-driven
H12b	E4q − rand-q, stft	+0.066 [+0.030, +0.100]	23/25	learned selection matters
H12b	E4q − rand-q, wpd	+0.071 [+0.037, +0.102]	22/25	learned selection matters
H12b	E4q − rand-q, wpds	+0.066 [+0.031, +0.099]	22/25	learned selection matters
H12c (desc.)	E4q − snr-q, stft	+0.020 [−0.006, +0.046]	18/25	descriptive
H12c (desc.)	E4q − snr-q, wpd	+0.013 [−0.013, +0.040]	17/25	descriptive
H12c (desc.)	E4q − snr-q, wpds	+0.007 [−0.021, +0.033]	10/25	descriptive
H11a	A_scat1 − A_stft	−0.026 [−0.058, +0.005]	4/25	NOT SUPPORTED
H11b	A_scat12 − A_stft	−0.013 [−0.044, +0.017]	7/25	NOT SUPPORTED
H11c	A_scat12 − A_scat1	+0.013 [+0.001, +0.025]	18/25	NOT SUPPORTED
H10a	ZCHSound clean any-CHD AUROC, W_stft	0.709 [0.675, 0.741]	n/a	SUPPORTED
H10b	AUROC(VSD) − AUROC(PFO)	+0.263 [+0.180, +0.343]	n/a	SUPPORTED
H10b	AUROC(VSD) − AUROC(ASD)	+0.126 [+0.064, +0.184]	n/a	SUPPORTED